

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

1**Water and solution properties**

COMP

You should be able to: Explain how the polarity and hydrogen bonding of water determine solubility and identify whether a given solute is hydrophilic or hydrophobic.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

2**pH and buffers**

COMP

You should be able to: Define pH, state the normal pH range of arterial blood, and explain how a buffer pair resists a change in pH when acid or base is added.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

3**Protein structure and function**

COMP

You should be able to: Relate the levels of protein structure to binding site shape and explain how denaturation by heat or pH change destroys function.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

4**Enzyme activity and regulation**

COMP

You should be able to: Describe how enzymes lower activation energy and predict the effect of substrate concentration, temperature, pH, and competitive or allosteric inhibition on reaction rate.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

5**ATP and energy coupling**

COMP

You should be able to: Explain how ATP hydrolysis is coupled to endergonic cellular work and name three categories of work that require it.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

6**Enzyme assay**

COMP

You should be able to: Measure enzyme activity across a range of temperature or pH using a spectrophotometric or colorimetric assay, plot the results, and identify the optimum.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____ ALSO A LAB SKILL

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

Every one of these boxes is **drawn**. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

7**ATP production pathways**

COMP

You should be able to: Compare glycolysis, the citric acid cycle, and oxidative phosphorylation by location, oxygen requirement, and ATP yield.

PROMPT ☐ A ☐ B PASS 1 COLOUR _____ PASS 2 COLOUR _____

1 · DRAW IT

a picture, not a paragraph

2 · LABEL EVERY PART

lines out to the drawing

3 · WHAT HAPPENS, IN ORDER

boxes and arrows, step to step

4 · WHAT BREAKS IF THIS FAILS

follow the arrows to what stops

Pass 1 from the book, before the video. Pass 2 in the second colour, after it, in the same boxes.

Dr. Sharilyn Rennie